

ActInf Textbook Group ~ Cohort 2 ~ Meeting 19 (Chapter 8, part 2)

TextbookGroup/ParrPezzuloFriston2022/Cohort_2 | Cohort 2 ~ Meeting 19 (Chapter 8, part 2) | 55:22

Hello, it is Cohort 2, March 8th, 2023. We're in meeting 19, and we're in our second discussion on chapter 8. We will look over the questions on chapter 8, and talk about any sections of chapter 8. And we will also probably take a preview look towards 9. So, just at the outset, does anyone have any comments or questions on 8? Anything else that they've come across or learnt over the last week that puts continuous time models in a different context? Any other thoughts on continuous and discrete? Or we'll look over the questions and make sure that we've at least touched upon each of the questions from the previous cohorts. Yeah, I'll let you go for it.

Actually, based on some of the claims, especially in a Bayesian mechanics paper, and also in this recent Path Integrals paper, they claim that FEP formulation, at least from about 2012 to 2019, was largely focused on developing density over states mathematics, or in other words, the first two types of FEP path tracking and path matching and path mode dynamics. But, I'm sorry, from that from that from that from that from that from that from that from that from that

they've concentrated more in developing the path integral formulation.

I'm not sure which direction this textbook leans toward,

because in some places it seems like it is the extension of the previous developmental trajectories.

But especially in Chapter 8, it gets more into this recent formulation of formulating the path integral mathematics.

But then again, even in Chapter 8, we see discussions about the...

Disappeared.

Discuss about the...

So, yes, I mean, previous years and this year, and also, yeah, how the textbook leans maps into that dichotomy.

Yeah, thank you.

Good points, Thomas, overviewed as such.

And...

This...

PAR textbook seems to be more heavily on the mode matching and mode tracking.

However, the continuous time...

It's kind of like the more things change, the more they stay the same.

So, looking at Thomas's overview of the history, the early implementations, as he noted, were in continuous time.

And then, in order to bring on explicit planning, discrete cognitive decision-making, symbolic state spaces,

and so on, sequential behavior,

they actually pivoted to the discrete time.

And that left the continuous time lock of Volterra, generalized dynamical systems, phrasings behind.

And then, now we have a fusion of the continuous and the discrete times, typified by the figure 8.6, where, through hierarchical nested models, we can include features of discrete decision-making, as well as continuous perception and action.

Also, discussed in Livestream 46,

ACNIMF does not contradict folk psychology with Kiefer, Ramsted, and Smith,

which discuss, essentially, this model as the folk psychological scaffold.

That there's a motor active inference layer that's perceptual and active in continuous space and time.

And then, we can propose these symbolic, discrete, explicit planning-oriented cognitive models, as cognitive models, I guess.

So, it's funny.

So, that's one return from the earliest, early 2000s continuous time.

Then, the focused move to discrete time.

And then, there's the hybrid models.

And then, as you pointed to, it even goes deeper,

because the discrete time formalism is perfectly suited for the first two phases of Bayesian mechanics.

And the non-equilibrium steady state and the state-space representations.

Whereas, for the truly path-tracking Bayesian mechanics,

and probably G-theory, chaos, and all of this,

continuous time is like your starter position

before it even goes further afield into gauge and fibers and sheaths

and all of these other formalisms.

So, it's almost like there's a sequence,

because the discrete time and the matrix multiplication

are very straightforward.

Then, when we look at figure 4.3 and figure 8.6,

we kind of see some similarities between continuous and discrete time.

And then, from the continuous time,

we can take another jump into the gauge setting,

which does seem to be the,

at least in principle,

the most modern representations we have of Bayesian mechanics.

Michael?

Yeah.

In figure 8.6,

we have in the top

the discrete time
and embedded
the continuous time.

I was wondering whether this can also be reversed.

We have in the top a continuous time
and then inside there,
we have sequences of discrete time.

So, for instance, if you,

I mean, we have a statistics
over how long a certain activity lasts,
and that can be broken down
into a sequence of several other activities.
could that be modeled in this way?

So, for instance,
for this overall activity,
we know that it takes,
I don't know,
two minutes or 20 seconds or so,
and then the next
action comes,
and each is broken down
into several smaller steps
that can be,
that are discrete,
can that be viewed in this way?

And then how,
for instance,
if we have a constraint over a time,
for instance,
there we have this,
what is this,
 τ_1 ,
this first state,
which could be maybe,
yeah,
if this was,
if one of those states
was constrained by a time,

and then this constraint
would be somehow handed down
to the embedded processes,
which each of which
could not take longer
than the constraint
in the state above.

Yeah, very interesting.

I've never seen
that,
such architecture,
but,
it's in principle,
possible.

It's very interesting.

And this kind of notion
of scheduling,
attentional scheduling,
and time constraints,
one way it might,
or kind of setting
where it could apply is,
this is, again,
brought up in the
folk psychology paper,
because the continuous time
lower level model
maps well to our
proprioception
and muscular action.

And then this
discrete time model
maps well to
sequential explicit planning.

So let's imagine
it was inverted.

The lower level model
might be a computer

doing a dispatch
in discrete time,
and the higher level
is like the human
operator.

And then they might
also have another
discrete above them.

So,
no need to stop
at just two layers.
I think it's very,
that's the exciting
direction in question,
is to build
these towers
of nested models,
and then
lateral connections
for collective
behavior
to
describe
and specify
the ecosystems
of shared
intelligence
so we can
understand how
different
causal
and cognitive
phenomena
arise from
interactions
within
and across
generative models.

Yeah.

And would there

be,

how could that

be modeled

if we have

a temporal

constraint

further down

in the

embedded

processes

which are

formulated

by the above

boxes

or states?

Is there

a way

to notify

the embedded

processes

that they should

stop eventually

or something

like this?

Yeah,

it's a...

I mean,

otherwise,

how is this

message passed

through?

The only

connection is

there are

these matrices

which is the

A matrix,
right?
And this
A matrix,
what they need
to determine
somehow
that the
embedded
processes
follow those
constraints.

Yeah.

Yes.

So let's
look at that.

So the
A matrix
is the
tail of
two densities.

So it can
be used as
a recognition
density
from the
observations
or a
generative
density
to generate
observations
from hidden
states,
just looking
at the
top.

A matrix

A matrix

A matrix

is a

Markov

blanket

between

S and

V.

Again,

Markov

blankets,

they don't

have anything

to do

a priori

with the

agent

environment

boundary,

though we

can identify

and construct

Markov

blankets

associated

with agent

boundaries.

This is

just using

Markov

blanket

to mean

the

node

or

nodes

upon

which
two
other
sets
of
states
are
conditionally
independent.
So the
O is
like the
interface,
it's a
generative
density,
speaking with
respect to
the higher
model,
it's a
generative
density
passing
observations
down
to
V,
the
slowly
varying
causes
which
modify
how the
derivatives
are constructed
in continuous

time.

So this

is kind

of like

a modifier

on the

Taylor

series

expansion.

And then

it is also

coming back

up,

thinking about

this in a

predictive

processing

like

bidirectional

way.

Also,

it's

changes

in the

continuous

time

setting,

lower

level

nested

model,

that

percolate

up and

modify

in a

perceptual

like way

higher
level
states.
So then
you ask
how could
constraints
be
modeled?
There
certainly
could
be
a
lot
of
engineering
type
solutions
like
just
sort
of
for
loops
and
virtualization,
emulation,
parallelization,
distributed
computational
processes,
dispatch
architectures,
a lot
of different
things that
aren't

clean
base
graph
only.
Just in
practice,
you would
end up
implementing
some of
these
strategies.
time
constraints.
But it's
interesting to
think about
how can
we really
in a
principled
way
include
time
constraints?
And then
when the
top level
is discrete,
how would
time
constraints
be
achieved?
And then
when the
top level
is

continuous,
you're
asking about
how would
time
constraints
be
achieved?

So
either way,
it's kind
of like
there's
like a
window,
like let's
just say
this is
the
real
time.

Here's
one more
dimension,
whether or
not the
top or
the bottom
is
continuous
or
discrete.

So this
could be
discrete,
discrete,
or
continuous,

continuous,
or any
combination.

One
frame
is
real
time
is
happening
on the
top.

And so
this is
like what
we've
seen in
sophisticated
inference,
sophisticated
affective
inference.

Here's
real
time.

This
is the
real
simulation
time.

1201,
1202,
1203.

And at
each
time point
in the
real

time,
there's
this
speculation
of the
past,
present,
and future.

Or
this
can be
the
real
time
model
in the
lower
model,
and the
higher
order
can be
speculative
planning.

or a
combination
where
there's
a
real
time
layer
that
has
both
a
faster
speculative

exploitation

layer,

as well

as a

slower

nested

planning

layer

above

real

time.

I

think

there's

many

disparate

things.

Once

we're

in the

space

of

nested

architectures

to

capture

complex

cognitive

phenomena,

it's a

really

open

space,

which

is why

it's so

important

to have

the
structured
learning
and the
statistical
power
testing
and the
ability
to
describe
and
understand
the
semantics
of these
models,
because I
think that
they're going
to get
out of
control
visually
very
quickly.
But
you're
saying
that there's
not only
information
going from
top to
bottom,
but also
the other
way around.

So the
embedded
processes
can inform
the upper
ones
about
some
outcomes.

Well,
yeah,
in principle,
so just
like O,
just only
looking at
the top
layer,
O via
A could
update
S,
or S
via A
could
produce
O.

That's
the
Taylor
two
density.

So the
Bayes
graph,
depending
on which
parameters

are
established
as
fixed
and
which
ones
are
established
as
learnable,
and so
on,
then
yes.
If you
allow
S to
be
learnable,
then
it will
learn
so that
it's
unsurprised
about the
incoming
observations
of
slowly
varying
causes.
Whereas
if you
fix
S
and you

only
allow
V
to be
learnable,
then
et cetera,
et cetera.
Hmm.
And that's
the kind
of
information
that's
actually
not
shown
on this
graph
other than
this line
which we
know is
missing.
But like
as
Toby
Sinclair
Smith's
dissertation
on
category
theory
has been
kind of
sharing
and
motivating,

there's
actually
information
that's
not
necessary
and
sufficient
to
replicate
a
full
model,
especially
if you're
going to
start
talking
about
learning
and
all
these
other
features.

So at
best,
this is
like a
quasi-formal
diagrammatic
mnemonic,
but not
quite the
necessary
and
complete
representation.

Okay.

But it's

a super

strong

motivator

and

helps us

think about

like,

you're a

great

question

about

continuous

time on

top and

we should

definitely

look into

that and

see if

those

models have

been used.

I think

part of it

is people

have just

seen a

natural

affinity

for

continuous

time

models

and

bodily

perception,

bodily
action.
discrete
time
models,
explicit
planning.
So
that's
just
commonly
held
perspectives.
So then
in that
situation,
it's
natural
to do
it
like
this.
But
it doesn't
mean it
can't be
otherwise.
Okay.
I'm not
sure if
these thumbs
up were
from our
own
time
last
okay.
This was

I think
from
office
hours.
We talked
about
figure
5.2,
predictive
coding,
message
passing,
and
8.4.
generalized
synchrony,
chaotic
systems.
Can
we
model
that
with
these
predictive
coding
architectures?
The
answer is
yes,
we can.
It just
depends on
how you
specify
it.
that
is

not
to
say
all
coupled
Lorenz
attractors
are
using
a
predictive
processing
architecture,
nor
that
predictive
processing
architectures
exhibit
Lorenz
attractor
behavior,
but
just
you
could
make
a
system
that
happens
to
have
both
of
those
features.
this

is
another
difference
between
the
continuous
and the
discrete
time
that I
don't
think
we've
emphasized.

Okay.

Action is
absent
from
equation
8.1.

This is
because
action
is part
of the
generative
process
not
the
generative
model.

Okay.

The
generative
model
only
deals
with

those
variables
directly
influenced
by
states
external
to
a
Markov
blanket.

But
those
are
only
the
sense
states
and
external
states
which
directly
influence
themselves
through
latent
causes.

But
doesn't
the
generative
model
also
include
cognitive
and
internal

states?

So one

!

So one

could argue

that the

realization

of action

or the

selection

of action

is part

of the

generative

process.

But can't

we say

that the

generative

model includes

planning as

inference?

is this

only in

the case

of the

continuous

time?

action?

So one

is

a

general

point

or if

this is

just referring

a little bit

to the
different
way that
action is
modeled in
the
continuous
time
setting
whereas
action in
the
discrete
setting
is modeled
in this
standard
policy
selection
way
 V
are
actually
like not
as much
even
described
as
policies
that can
be selected
because it
doesn't make
as much
sense to
talk about
counterfactuals
on V

they're
described as
just slowly
varying causes
in the
generative
process
basically
contingencies
or dependencies
upon which
the Taylor
series
approximation
is impinged
on by
that's
like a
very
oblique
way of
talking about
the
consequences
of action
it's
almost
like a
differential
angle
on
action
Michael
yeah so
I'm
wondering
whether
this

Taylor
series
decomposition
this
different
derivations
that you
have
there
whether
this is
only one
view
on
continuous
time
series
or whether
there's
also other
views
that do
not imply
a derivation
of
decomposition
of one
function
into
several
derivations
I'm not
sure
if I
understand
this
correctly
but

couldn't it
be that
time
constraints
are
related
without
derivations
I'm just
wondering
whether
this is
only one
way of
seeing it
or
is it
the only
one
yeah good
question
yeah it's
not exactly
a Taylor
series
I think
it's
though it
has a Taylor
series like
nature
because it
is dealing
with successive
derivative
approximations
with their
own

coefficients
but I
think the
more physics
framing is
the generalized
coordinates
so position
momentum
acceleration
jerk
etc
is there
a way to
talk about
time
constraints
and multiple
time scales
in continuous
time
other than
by taking
derivatives
I mean we
can definitely
explore it
but like the
first thought
that comes to
mind is
when faced
with a
field of
continuous
time
two approaches
that are

taken
one of
them is
to
discretize
time
at which
point you'd
be using
a discrete
time model
and if
you're not
going to
discretize
time
you're
going to
leave it
as a
smooth
continuum
then a
general
approach
is to
utilize
physics
based
generalized
coordinates
of motion
Ali
well yeah
actually
Michael
Silverstein
and

William
Stocky
has been
arguing
for another
approach
to
formalize
cognition
time
and
other
so-called
dynamical
aspects of
the system
using
adynamical
mathematics
so for
instance
in their
paper
rethinking
the world
with
neutral
monosome
removing
the boundaries
between
mind
matter
and
space
time
they've
proposed

an
alternative
approach
which is
somehow
which is
expanded
more
I mean
in detail
in their
Beyond the
Dynamical
Universe
book
but
in this
case
they
so very
briefly
they
conceptualize
a system
as
a kind
of a
stationary
system
that
just
moves
from
sorry
that just
has
the first
initial

position
and
the
ending
ultimate
position
and
we can
by
using
this kind
of
adynamical
mathematics
remove
the
time
from
the
equation
and
just
actually
look at
the
system
in a
kind
of
block
universe
way
without
actually
taking
into
consideration
the

flow
of
the
system
through
time
so
yeah
there
is
definitely
other
approaches
to
model
these
kinds
of
systems
and
even
they
argue
in this
paper
and
also
in
their
book
that
this
new
approach
can
somehow
overcome
some

of the
long-standing
issues
in
theoretical
physics
and
or even
in the
philosophy
of
physics
yeah
very
interesting
this
kind
of
well
writing
out the
ontological
reality of
time in
the world
is definitely
a complex
topic
so it's
always good
to know
that with
these
discussions
about
time
we're
talking

about
map
time
about
like how
we want
to
model
time
and in
that
setting
both
8.1
and 8.2
have

a
timeless
data
generating
function
y
observations
is simply
a function
of latent
states
on hidden
states
so
y
observational
states
is a
function
of latent
states

x
plus
some
sensor
variance
y
and then
there's
a
field
or
space
in which
latent
states
have
their
own
flow
that
is
defined
again
without
reference
to
time
but
as
a
derivative
field
in a
Hamiltonian
sense
so
even
if

your
y
data
stream
is
an
EEG
or
fMRI
time
series
data
it's
actually
like
where's
the
time
oh
where
does
the
time
go
not
seeing
any
t's
in
these
continuous
time
models
so
it's
almost
like
we

lean
so
hard
into
the
continuity
of
time
and
continuity
is
at least
synonymous
with
or
adjacent
to
differentiable
those
two
they're
not
exactly
the
same
but
we
work
with
continuous
functions
because
we
want
them
to
be
differentiable

we
have
to
discretize
them
but
the
biggest
issue
with
discrete
functions
is
that
they're
non
differentiable
Ali
I
think
the
more
the
more
the
mathematical
condition
for
modeling
these
systems
is that
they need
to be
in
wiener
spaces
so

wiener
spaces
so
wiener
spaces
is
just
a
generalized
space
that
contains
all
the
continuous
and
differentiable
functions
and
also
I just
wanted to
point it
out
that
that's
exactly
the
point
or at
least
one of
the
points
of
using
generalized
coordinate

systems
in
path
integral
formulation
of
FEP
by
I mean
by removing
the
time
from
the
equations
it
is
possible
to
talk
about
the
behavior
of
the
system
purely
in
terms
of
their
space
of
the
space
of
possible
paths

instead
of
just
looking
at
instead
of
just
following
the
time
developmental
aspect
of
the
particular
states
awesome
yeah
it's
like
the
difference
between
putting
a drop
of
food
coloring
in the
cloud
and
tracing
that
one
path
as it
flows

and
then
there's
definitely
there's
this
discussion
oh
should
we
model
that
one
path
of
the
tracer
as a
path
with
continuous
state
space
or
should
we
discretize
it
and do
a
discrete
time
model
of that
one
realized
trace
or

should
we
step
back
describe
the
topology
of
the
mountain
and
then
the
realized
trace
will be
a path
of least
action
on that
landscape
but
we'll
also
be able
to talk
about
counterfactuals
and
alternative
starting
conditions
initial
conditions
and boundary
conditions
and
and

and ask
what if
questions
on the
landscape
whereas
any
descriptive
historical
model
will not
natively
or simply
enable
counterfactual
comparisons
if you just
have the
path
descriptor
in
continuous
time
or in
discrete
time
then
you don't
even know
what the
adjacent
possibles
are
slash
could
have
been
Michael

yeah
I'm
I'm
I'm
not sure
I understand
this
is
is this
possible
because
we assume
that
the
the time
goes
in regular
intervals
so
if it's
a regular
interval
we can
just
ignore
it
or
what
is
the
underlying
idea
to
obstruct
away
from
time
because

time
seems
such a
crucial
kind
of
element
in
all
our
daily
lives
right
yeah
go for
it
well
yeah
it
definitely
is an
essential
component
but
I mean
the point
here is
that
you see
so for
instance
in classical
mechanics
we have
at least
three
different
formulations

one is
the
Newtonian
mechanic
formulations
the other
one is
the
Lagrangian
and also
the
Newtonian
formulation
definitely
needs
time
as an
explicit
variable
and parameter
but
going
from
Newtonian
formulation
to
Lagrangian
and
Hamiltonian
especially
in the
Hamiltonian
formulation
we can
rearrange
or derive
those
formulations

from
the
classical
Newtonian
formulation
without
accounting
for
time
as an
explicit
variable
so
time
is
actually
implicit
in
those
formulations
so
all three
of those
formulations
are equivalent
and they
can be
recovered
from each
other
and they
can be
derived
from each
other
but
it's
just a

matter
of
which
parameters
are
explicit
and
which
are
implicit
so
yeah
we
don't
get
rid
of
time
entirely
we
just
use
time
as
a
kind
of
implicit
parameter
that
doesn't
need
to be
explicitly
defined
yeah
very
interesting

and also
you said
the assumption
time is in
regular
intervals
well that's
a discrete
time
model
and that's
the question
of
discretization
of time
which we're
actually
like exploring
in
live stream
53
which is
again not
the ontological
reality
of the
time
the time
click of
the world
or whatever
but in
modeling
spaces
how to
retain
the
elegance

and the
analytical
advantages
that
continuous
models
provide
in
terms of
generalized
coordinates
of
motion
physics
grounded
energy
based
learning
convergence
guarantees
around free
energy
minimization
distribution
those

are all
about
continuous
differentiable
spaces
which is
why they're
grounded in
information
geometry
and functional
analysis

whereas
discretization
we want to
make sure
that the
discretization
techniques
that are
used
don't
solely
or
lose
those
properties
of the
underlying
continuous
distribution
information
michael
yeah
i'm
i'm digesting
this
it's a bit
so
so
so
so
actually
derivation
is
takes the
takes the
role of
time
so it's only

derivable
because we
assume there
is some
kind of a
continuation
or some
kind of
time
underlying
yeah
let's
pull back
to a
description
so
if we
wanted a
time
description
of a
car
we could
describe its
position at
every time
and so
arguably one
could say
well that
described
like
everything
about that
car's
trajectory
what you
need to

know is
its position
at every
time
all right
well you
could also
use the
generalized
coordinates
of motion
and so
for each
time
you could
also have
the car's
position
in x
and y
but also
you could
have the
car's
acceleration
and and
and so
on
so those
are the
derivatives
of the
position
and
it's not
like
extra
information

like with
the whole
trajectory
you could
recalculate
the generalized
coordinates
of motion
from the
position
but you
also might
want your
data structure
to be
that vector
at each
time step
with the
generalized
coordinates
of motion
so you
know it's
here and
it's going
50 kilometers
an hour
and it's
slowing down
by 2
kilometers
per hour
per second
and its
acceleration
is changing
by you

know
positive
1 kilometer
per second
squared
and so
on
another
angle
would be
to
describe
that
that
space
upon
which
there
are
trajectories
of the
generalized
coordinates
now there's
some
generalized
coordinates
locations
that are
just
ridiculous
like
well it's
not moving
at all
but its
fourth derivative
is super

high
well
any state
space
can be
used to
describe
certain
points
in that
state
space
that
aren't
like
feasible
or
you know
realizable
by a
different
physical
system
obviously
systems
spend
their
time
in
sometimes
highly
constrained
subspaces
manifolds
of these
bigger
state
spaces

but
when you
say
okay
well
then let's
make the
generalized
coordinates
of motion
our
fundamental
state
space
so instead
of
xyz
t
as our
fundamental
state
space
and then
that is
and then
we're going
to derive
all of our
other
questions
from there
it's like
saying
well let's
have
the generalized
coordinates
be our

state
space
but then
you don't
need
t
of course
real
trajectories
unfold
in
quote
time
pending
future
understandings
of what
time
is
but
you don't
need
t
in the
Hamiltonian
which
again
just to
connect
back to
the
textbook
is
that
is what
we're
seeing
in these

representations

that

basically

partitions

signal

from

noise

without

explicit

reference

to

t

as a

variable

however

amenable

to

fitting

empirical

time

series

data

I don't

think we

resolve

this

question

about

action

being

in

the

generative

process

what

exactly

is

meant

by
that
because
action
itself
is part
of
the
generative
process
we
could
say
the
consequences
of
action
or
the
embodiment
of
but
is
this
claim
about
generative
models
in
general
or is
this
just
saying
something
about
continuous
time

okay
precision
!
in

in
our
office
hours
last
week
we
had
some
good
conversations
about
precision
and

more
more

more
!
and
pain
some
more
!

the
bird
song

can perspective active inference precision and attention are synonyms
here was the discussion from um the office hours so this is a it's related to chapter eight
so we were talking about um okay so in the examples first we have just general lock of
winterless competition um that is then deployed in a handwriting example showing how lock of
like models can account for sequential and reactive behavior without explicit planning
so in the discrete time it'd be like i'm here i'm likely to go
whereas in um the continuous time setting it's just like a dynamical attractor
that doesn't have an explicit representation of where it's going to go in the attractor
it's just flowing in the attractor and so then we describe the state space of the attractor
the attractor

[illegible]

hmm wow thank you yeah i haven't read this one in depth this day in in the blockference we uh
so there is there was the original paper there was a guest stream with brett kagan

and uh adil razzi on the dish brain and of course the uh the sentience furor that arose
leading to um multiple and apparently ongoing sequences of short letters
there's a lot of interesting stuff in here
the big contention is with the terms act in fontology sentience goal-directed behavior
embodiment uncertainty intelligence and their call is um for glossaries
and then this paper also
all right come develop the glossaries
but this was just they're in the block i'll put it in the chat as well for people who want to see that
um
like this meta on being able to separate the technical
role it's not just map territory but i know we bring it up all the time and it comes up again and again
but or maybe it is all just simply that issue
or just something about using natural terms
especially day-to-day terms
to refer to technical parts or outcomes or descriptors of models
so it's all good to just accept this and say well it's an assertion by the authors
just verbally
that
uh dollar sign term one and dollar sign term two are synonymous
that's all good to just
have as a neutral rhetorical assertion
but if we want to start latching that into our understanding of the world
then
seems like we need to
i don't know
michael just added you to the um
document seal boxes there
all right let's um in our closing few minutes here
i think we talked about eight a fair amount
also eight is not that long
slash not that
of course one could go very deep
into
the mathematics of the
continuous
or the hybrid models but we
we um
and we could go into the other examples and so on but um

you know

just first pass there

okay so that is the end of chapter eight

let's just quickly in the last few minutes look ahead to nine

just because we have the best hammer does not mean that every problem is a nail

models are useful if they can answer scientific questions

chapter nine focuses on active inference in the context of empirical data

section 9.2 is on metabasean methods

and it's going to talk about the setting of

cognitive modeling with active inference

in the setting of behavioral or ethological experimentation

and so it's not just going to be a proof of principle

like

we constructed a t-maze and a simulated rat

um

there's two reasons

to fit a computational model to observed behavior

the first reason is to estimate parameters of interest

the second reason is to compare alternative hypotheses

they recap bayes theorem and they're going to be discussing bayes theorem in the context of
scientists scientific epistemology

and a really bayesian epistemology

as the behavioral investigator

that

lends itself to this key figure 9.1

that characterizes the metabasean view

so in the inner dashed box we have the cognitive model that we're constructing of the ant

so map territory

level one

then we're in our own dashed box

because we are providing what for us are action selections and parameters of the experiment

drawn from distributions of experiments and stimuli

and presenting it to the ant

then potentially with errors and variance etc. observing behavior

so this is the metabasean view

view from the outside

and view from the inside

now there is a view from the inside

but this is a map of the view of the inside
and this is our schema
of our control framework from the outside
so all the talking about
sweeping over families of cognitive models
of ants and rats and stuff like that
that's all happening in this dashed box
but in the outer loop
we are exploring different maximally informative
cognitive modeling settings
and observations to observe
and again like the imperative being maximally informative information through experiments
not some kind of falsificationist
accept or reject dialectic
but putting it more into a base factor
continuous
juxtaposition
of portfolios of hypotheses
rather than like a p-value frequentist
accept or reject framework
it's going to be like
of active inference
in multiple ways
okay
turn away from that sort of ethological view
to some technical details
section 9.4
parametric empirical bays
PEB
is a way that you can start
with a real data set
and kind of use the first instantiation
of the whole data set
or a part of the data set
to jump start you
or like
parachute drop you
into

a reasonable area
of phase space
and that facilitates
expectation maximization type algorithms
and
empirical applications
of Bayesian statistics
like
we're going to fit our prior
on this 5th grade classroom's heights
from
a bigger data set
of heights that were collected
from 1st through 8th grade
and that's going to be our prior
for this 5th grade classroom
or we're just going to take
the measurements we did get
and then summary statistics
of that distribution
we're going to use
as our prior
then 9.5
it's kind of like a little bit
like a chapter 6
a recipe
but here the recipe
is about collecting data
formulating a model
as in chapter 7
or in chapter 6
if you want to pull back
even more generally
specify likelihood
priors
solve for
probabilities
and evidences

and then perform
statistical analyses
and that's actually where
descriptive statistics
can come into play
like
we did in ANOVA
on these cognitive
modeling parameters
and then
they have a
summary figure

9.2

so here we're like
really bringing together
a lot of sections
of the book
in slide 1
is the TMAs
here's the generative
process
and generative model
that could have been
written before
lands a likelihood
function
that allows
this invertible model
such that you can
use
the generative model
to generate
synthetic data
or
you can
take data
of that structure
and do

statistical inference

on

so we did

five conditions

and it turns out

condition 4 and 5

with this p-value

are significantly higher

or this base factor

than there

and

some examples

of generative models

some models

of false inference

a long table

and a summary

all right

thank you

that

concludes

chapter 8

we'll

return in one minute

or two minutes

with

beginning of

next cohort

thank you

you